

KINGSTON MANLEY, Jamaica

WMO: 783970

Lat: 17.936N

Lon: 76.788W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.9	73.5	58.6	73.4	77.5	60.5	78.7	77.9	30.0	85.0	27.6	84.8	6.5	350	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	91.8	79.0	91.3	79.1	90.1	78.7	82.6	87.1	81.8	87.0	81.0	87.0	20.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.6	164.5	84.9	80.6	159.1	84.7	79.3	152.1	84.9	46.5	86.6	45.5	87.7	44.8	87.0	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.2	25.5	23.3	DB	69.8	94.7	2.6	1.8	67.9	96.0	66.4	97.1	65.0	98.1	63.1	99.4	
			WB	62.7	84.4	2.4	1.3	61.0	85.3	59.6	86.1	58.3	86.8	56.6	87.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	80.4	80.2	80.8	82.2	83.4	84.8	85.2	85.1	84.7	83.7	82.7	81.3
	DBStd	2.67	1.91	1.75	1.86	1.71	1.86	2.14	1.89	1.95	2.07	2.18	1.97	2.19
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12004	942	845	955	967	1036	1045	1091	1087	1041	1045	980	970
	CDD65	6529	477	425	490	517	571	595	626	622	591	580	530	505
	CDH74	73998	4350	3861	4730	5633	6746	7576	8049	7953	7351	6843	5849	5057
	CDH80	27295	1097	952	1250	1775	2505	3356	3654	3571	3141	2612	1926	1456
Wind	WSAvg	9.8	8.8	9.3	9.4	9.7	11.1	13.8	12.3	10.7	9.0	8.1	7.4	8.2
Precipitation	PrecAvg	28.6	0.7	0.5	0.7	1.1	3.1	2.7	1.4	3.5	4.8	5.8	3.1	1.5
	PrecMax	67.1	4.1	2.2	2.9	5.2	12.2	21.6	6.5	21.6	21.5	25.0	13.7	7.7
	PrecMin	7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0
	PrecStd	13.4	0.8	0.4	0.7	1.2	3.4	3.8	1.6	3.8	5.4	5.7	3.0	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.5	89.2	89.2	89.9	91.2	93.2	93.4	93.0	92.9	91.9	91.3	90.0
		MCWB	74.7	75.6	75.7	77.1	78.6	78.0	78.6	79.5	80.1	79.4	78.6	76.9
	2%	DB	87.7	87.5	87.7	88.2	89.7	91.5	91.7	91.5	91.3	90.4	89.6	88.2
		MCWB	75.0	75.5	75.8	77.0	78.4	78.8	78.7	79.7	79.7	79.1	78.0	76.5
	5%	DB	86.2	86.1	86.3	87.7	88.5	90.0	90.6	90.0	89.9	89.5	88.2	87.5
		MCWB	74.8	75.0	75.4	76.9	78.1	78.6	78.6	79.3	79.3	78.9	77.7	76.2
	10%	DB	85.0	84.6	85.6	86.3	87.8	89.4	89.6	89.5	89.3	88.0	87.4	86.1
		MCWB	74.3	74.2	75.1	76.1	78.0	78.5	78.5	79.1	79.1	78.6	77.4	75.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	79.9	80.0	80.9	82.6	82.8	82.7	83.6	83.6	82.7	82.3	80.4
		MCDB	83.2	84.8	84.5	86.3	86.9	87.1	87.5	88.1	87.3	86.6	85.7	84.8
	2%	WB	77.6	77.8	78.4	79.3	80.8	81.4	81.5	82.2	82.2	81.5	80.7	79.0
		MCDB	83.2	84.1	83.8	85.4	86.7	87.4	87.7	87.6	86.7	86.4	85.1	84.6
	5%	WB	76.6	76.6	77.3	78.4	79.9	80.5	80.5	81.3	81.3	80.5	79.6	77.9
		MCDB	83.0	83.4	83.6	84.8	86.1	87.6	87.9	87.5	86.9	86.3	85.0	84.2
	10%	WB	75.6	75.5	76.3	77.5	79.1	79.8	79.7	80.3	80.4	79.7	78.7	77.0
		MCDB	82.6	82.6	83.1	84.4	85.6	87.1	87.4	87.2	86.9	85.8	84.9	83.8
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.5	10.1	9.4	9.1	9.0	9.6	9.4	9.4	9.5	9.6	10.2
		MCDBR	11.6	11.6	11.0	10.2	9.7	9.8	10.3	9.9	10.0	10.5	10.5	11.3
		MCWBR	6.3	6.6	6.3	5.9	5.7	5.8	6.2	6.1	6.1	5.8	5.8	6.0
	5% WB	MCDBR	10.9	11.0	10.4	9.5	9.1	9.0	9.5	9.3	9.6	9.9	9.9	10.5
MCWBR		6.7	7.0	6.5	6.2	6.0	5.9	6.5	6.4	6.4	5.9	6.2	6.5	
Clear-Sky Solar Irradiance	taub	0.386	0.388	0.391	0.425	0.455	0.507	0.509	0.472	0.431	0.395	0.392	0.383	
	taud	2.519	2.497	2.485	2.399	2.341	2.171	2.157	2.299	2.429	2.552	2.532	2.550	
	Ebn at Noon	285	290	292	280	268	252	252	264	276	285	281	283	
	Edh at Noon	32	34	36	39	41	48	49	43	37	32	31	30	
All-Sky Solar Radiation	RadAvg	1437	1676	1873	2006	1901	1912	1993	1902	1791	1547	1425	1343	
	RadStd	76	61	75	83	124	112	123	96	135	139	98	69	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page